

SHACHE, China

WMO: 518110

Lat: 38.433N

Lon: 77.267E

Elev: 4042

StdP: 12.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.7	10.7	-2.6	5.5	18.1	1.5	6.9	20.9	7.8	32.5	7.3	31.4	1.4	270	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	94.3	65.3	91.8	64.5	89.5	63.9	71.2	85.1	69.1	83.9	67.4	83.1	4.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	115.6	79.3	64.4	105.2	77.2	62.2	97.4	75.9	38.0	85.0	35.9	84.2	34.4	82.9	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.3	7.7	4.8	99.2	6.0	2.0	0.4	100.6	-3.1	101.8	-6.5	102.9	-10.9	104.3		
(5)			4.0	74.6	5.9	3.7	-0.3	77.2	-3.7	79.4	-7.0	81.5	-11.2	84.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	23.0	33.3	49.3	61.7	69.1	75.7	78.1	75.7	67.8	55.8	41.4	27.3		
	DBStd	19.77	6.49	6.86	7.38	6.61	5.57	4.72	4.50	4.31	4.51	5.87	6.51	5.41		
	HDD50	2401	837	468	106	7	0	0	0	0	0	14	267	702		
	HDD65	5028	1302	888	488	133	27	1	0	1	23	289	709	1167		
	CDD50	4209	0	0	84	358	593	771	872	796	533	194	8	0		
	CDD65	1362	0	0	1	34	154	322	408	332	107	4	0	0		
	CDH74	13833	0	0	16	466	1520	3340	4333	3104	977	77	0	0		
	CDH80	5545	0	0	1	114	478	1433	2004	1264	247	4	0	0		
(14) Wind	WSAvg	3.0	2.2	2.7	3.6	4.0	4.0	4.1	3.5	3.2	2.8	2.3	2.1	2.0		
(15) Precipitation	PrecAvg	1.9	0.0	0.1	0.1	0.2	0.4	0.3	0.2	0.3	0.1	0.1	0.1	0.0		
	PrecMax	4.9	0.2	0.5	1.1	1.7	1.7	1.6	0.9	2.4	0.7	1.2	0.8	0.2		
	PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.2	0.1	0.1	0.2	0.4	0.5	0.4	0.2	0.5	0.2	0.2	0.2	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.1	56.6	76.1	87.9	90.6	95.9	98.4	95.9	88.7	79.8	68.3	50.6		
		MCWB	34.2	39.7	49.7	56.9	59.9	64.2	67.4	68.5	63.1	55.9	46.8	36.6		
	2%	DB	40.4	52.4	71.0	83.4	87.9	93.0	95.3	91.9	85.6	76.1	62.3	43.1		
		MCWB	30.1	38.0	47.4	55.0	58.7	62.5	66.1	66.4	61.6	55.0	44.3	33.3		
	5%	DB	36.4	48.7	67.3	79.5	85.4	90.5	92.6	89.4	82.8	72.6	58.1	39.1		
		MCWB	28.1	36.1	46.0	53.6	58.5	62.0	65.9	66.5	60.8	53.6	42.7	30.7		
	10%	DB	33.0	45.0	63.5	75.7	82.5	88.0	89.9	87.0	79.9	69.0	53.7	35.8		
		MCWB	26.0	34.0	44.5	51.8	57.4	61.5	65.2	65.9	60.0	51.3	40.9	28.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.7	42.3	50.6	58.4	65.3	69.3	74.7	74.6	68.6	59.2	48.1	37.4		
		MCDB	47.1	53.6	73.4	82.5	82.5	85.0	86.9	88.0	81.7	73.3	65.0	49.3		
	2%	WB	30.6	39.0	48.3	56.5	62.2	66.5	71.5	71.5	65.1	56.7	45.3	33.6		
		MCDB	39.7	50.2	68.7	79.3	80.5	83.6	85.3	85.1	78.9	72.1	60.6	42.0		
	5%	WB	28.3	36.6	46.5	54.7	60.3	64.6	69.3	69.2	63.2	54.9	43.2	31.2		
		MCDB	35.8	47.4	65.1	76.0	79.4	83.7	84.7	83.2	76.7	69.8	56.1	38.4		
	10%	WB	26.4	34.4	45.0	53.0	58.9	63.3	67.8	67.5	61.6	52.9	41.3	29.2		
		MCDB	32.4	44.2	61.7	72.9	78.4	82.7	84.2	82.2	75.1	66.2	53.0	35.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	18.1	20.2	21.9	22.0	21.8	21.1	20.4	22.3	25.6	22.6	16.5		
		MCDBR	22.7	23.0	25.5	29.0	27.1	25.6	25.3	24.9	27.1	28.3	29.2	21.0		
	5% WB	MCWBR	14.6	12.5	11.1	11.0	10.5	9.3	9.8	10.5	10.9	12.2	14.4	12.9		
		MCWBR	21.5	21.8	24.1	25.9	24.1	22.8	22.6	22.8	24.4	26.4	28.4	20.2		
(40) Clear-Sky Solar Irradiance	taub		0.355	0.417	0.546	0.607	0.618	0.610	0.601	0.652	0.632	0.520	0.432	0.381		
	taud		2.060	1.924	1.642	1.551	1.584	1.632	1.674	1.578	1.573	1.727	1.878	1.986		
	Ebn at Noon		249	246	224	221	222	224	225	208	200	209	215	226		
	Edh at Noon		38	50	74	86	85	81	77	82	78	61	46	38		
(44) All-Sky Solar Radiation	RadAvg		824	1105	1522	1832	2127	2259	2143	1894	1650	1371	991	771		
	RadStd		121	105	97	65	117	94	119	125	96	51	49	75		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.81	N/A	N/A	+0.94	-2.48	-3.74	N/A	-202	+241	+124
(47) Regional (3 neighbors)		+0.58	N/A	N/A	N/A	-1.87	-2.27	N/A	N/A	+218	+97

Nomenclature: See separate page